

BFO 2020 Spatial Axioms

Occurs in and environs are inverse relations [uys-1]

$$\forall a,b(\text{occursIn}(a,b) \leftrightarrow \text{environs}(b,a))$$

Located in and location of are inverse relations [kaw-1]

$$\forall t,a,b(\text{locatedIn}(a,b,t) \leftrightarrow \text{locationOf}(b,a,t))$$

Occurs in is disjunctive on first argument when it is an occurrent [jil-1]

$$\forall p,q,r(\text{occursIn}(p,q) \wedge \text{occurentPartOf}(r,p) \rightarrow \text{occursIn}(r,q))$$

If a process (or process boundary) occurs in a continuant, that continuant exists at least as long as the process does [dxv-1]

$$\forall p,c(\text{occursIn}(p,c) \rightarrow \forall t(\text{existsAt}(p,t) \rightarrow \text{existsAt}(c,t)))$$

Located in is disjunctive on third argument, a temporal region [put-1]

$$\forall p,q,r,s(\text{locatedIn}(p,q,r) \wedge \text{temporalPartOf}(s,r) \rightarrow \text{locatedIn}(p,q,s))$$

Located in is a lower bound on second argument [evu-1]

$$\forall p,q,r,s(\text{locatedIn}(p,q,r) \wedge \text{continuantPartOf}(q,s,r) \rightarrow \text{locatedIn}(p,s,r))$$

Located in is disjunctive on first argument when it is a continuant [wtv-1]

$$\forall p,q,r,s(\text{locatedIn}(p,q,r) \wedge \text{continuantPartOf}(s,p,r) \rightarrow \text{locatedIn}(s,q,r))$$

Occupies spatial region is functional on second argument [zls-1]

$$\forall p,q,r,s(\text{occupiesSpatialRegion}(p,q,r) \wedge \text{occupiesSpatialRegion}(p,s,r) \rightarrow q=s)$$

Occupies spatial region is functional on first argument [csx-1]

$$\forall p,q,r,s(\text{occupiesSpatialRegion}(p,q,r) \wedge \text{occupiesSpatialRegion}(s,q,r) \rightarrow p=s)$$

Occupies spatial region is disjunctive on third argument, a temporal region [mud-1]

$$\forall p,q,r,s(\text{occupiesSpatialRegion}(p,q,r) \wedge \text{temporalPartOf}(s,r) \rightarrow \text{occupiesSpatialRegion}(p,q,s))$$

Spatially projects onto is disjunctive on third argument, a temporal region [ivt-1]

$$\forall p,q,r,s(\text{spatiallyProjectsOnto}(p,q,r) \wedge \text{temporalPartOf}(s,r) \rightarrow \text{spatiallyProjectsOnto}(p,q,s))$$

Located in is transitive at a time [ets-1]

$$\forall a,b,c,t,t2(\text{locatedIn}(a,b,t) \wedge \text{locatedIn}(b,c,t2) \wedge \text{temporalPartOf}(t,t2) \rightarrow \text{locatedIn}(a,c,t))$$

If a location of b then if a is an instance of continuant fiat boundary then b is an instance of continuant fiat boundary [wte-1]

$$\forall p,q,t(\text{locationOf}(p,q,t) \wedge \text{instanceOf}(p,\text{continuantFiatBoundary},t) \rightarrow \text{instanceOf}(q,\text{continuantFiatBoundary},t))$$

If a process occurs in c then it occurs in any place that c is part of at least as long as p exists [czc-2]

$$\forall p,c1,c2(\text{occursIn}(p,c1) \wedge (\forall t(\text{existsAt}(p,t) \rightarrow \text{continuantPartOf}(c1,c2,t))) \rightarrow \text{occursIn}(p,c2))$$

All spatial regions are part of a 3 dimensional spatial region [xcx-1]

$$\forall s,t(\text{instanceOf}(s,\text{spatialRegion},t) \rightarrow \exists s3(\text{instanceOf}(s3,\text{threeDimensionalSpatialRegion},t) \wedge \text{continuantPartOf}(s,s3,t)))$$

Spatial regions don't change what they are part of. [mlb-2]

$$\forall sr1,sr2,t1,t2(\text{continuantPartOf}(sr1,sr2,t1) \wedge (\text{existsAt}(sr2,t2) \vee \text{existsAt}(sr1,t2)) \wedge \text{instanceOf}(sr2,\text{spatialRegion},t1) \rightarrow \text{continuantPartOf}(sr1,sr2,t2))$$

Material entities occupy three dimensional spatial regions [hpx-1]

$$\forall m,t(\text{instanceOf}(m,\text{materialEntity},t) \rightarrow \exists t1,s(\text{temporalPartOf}(t1,t) \wedge \text{instanceOf}(s,\text{threeDimensionalSpatialRegion},t1) \wedge \text{occupiesSpatialRegion}(m,s,t1)))$$

If something is located in something else then the region of the first is part of the region of the second [uas-1]

$$\begin{aligned} &\forall a,b,t (\text{locatedIn}(a,b,t) \\ &\quad \rightarrow \exists r_1,r_2,t_2 (\text{temporalPartOf}(t_2,t) \wedge \text{occupiesSpatialRegion}(a,r_1,t_2) \\ &\quad \wedge \text{occupiesSpatialRegion}(b,r_2,t_2) \wedge \text{continuantPartOf}(r_1,r_2,t_2))) \end{aligned}$$

Occurs in has domain process or process boundary and range material entity or site [tfw-1]

$$\begin{aligned} &\forall a,b (\text{occursIn}(a,b) \\ &\quad \rightarrow (\exists t (\text{instanceOf}(a,\text{process},t) \vee \text{instanceOf}(a,\text{processBoundary},t))) \\ &\quad \wedge (\exists t (\text{instanceOf}(b,\text{materialEntity},t) \vee \text{instanceOf}(b,\text{site},t)))) \end{aligned}$$

Occupies spatial region is time indexed and has domain: independent continuant but not spatial region and range: spatial region [lzw-1]

$$\begin{aligned} &\forall a,b,t (\text{occupiesSpatialRegion}(a,b,t) \\ &\quad \rightarrow \text{instanceOf}(a,\text{independentContinuant},t) \wedge \neg \text{instanceOf}(a,\text{spatialRegion},t) \\ &\quad \wedge \text{instanceOf}(b,\text{spatialRegion},t) \wedge \text{instanceOf}(t,\text{temporalRegion},t)) \end{aligned}$$

If there are two independent continuants that are not spatial regions, and one is part of the other, then it is located in the other [bao-1]

$$\begin{aligned} &\forall a,b,t (\text{continuantPartOf}(a,b,t) \wedge \text{instanceOf}(a,\text{independentContinuant},t) \\ &\quad \wedge \neg \text{instanceOf}(a,\text{spatialRegion},t) \wedge \text{instanceOf}(b,\text{independentContinuant},t) \\ &\quad \wedge \neg \text{instanceOf}(b,\text{spatialRegion},t) \\ &\quad \rightarrow \text{locatedIn}(a,b,t)) \end{aligned}$$

Spatial region is the union of zero dimensional spatial region, one dimensional spatial region, two dimensional spatial region, and three dimensional spatial region [wnm-1]

$$\begin{aligned} &\forall i,t (\text{instanceOf}(i,\text{spatialRegion},t) \\ &\quad \rightarrow \text{instanceOf}(i,\text{zeroDimensionalSpatialRegion},t) \\ &\quad \vee \text{instanceOf}(i,\text{oneDimensionalSpatialRegion},t) \\ &\quad \vee \text{instanceOf}(i,\text{twoDimensionalSpatialRegion},t) \\ &\quad \vee \text{instanceOf}(i,\text{threeDimensionalSpatialRegion},t)) \end{aligned}$$

No two material entities occupy the same space unless they coincide [scr-1]

$$\begin{aligned} &\forall m_1,m_2,s,t (\text{instanceOf}(m_1,\text{materialEntity},t) \wedge \text{occupiesSpatialRegion}(m_1,s,t) \\ &\quad \wedge \text{instanceOf}(m_2,\text{materialEntity},t) \wedge \text{occupiesSpatialRegion}(m_2,s,t) \\ &\quad \rightarrow (\text{continuantPartOf}(m_2,m_1,t) \wedge \text{continuantPartOf}(m_1,m_2,t)) \vee m_1=m_2) \end{aligned}$$

Located in is time indexed and has domain: independent continuant but not spatial region and range: independent continuant but not spatial region [bge-1]

$$\begin{aligned} &\forall a,b,t (\text{locatedIn}(a,b,t) \\ &\quad \rightarrow \text{instanceOf}(a,\text{independentContinuant},t) \wedge \neg \text{instanceOf}(a,\text{spatialRegion},t) \\ &\quad \wedge \text{instanceOf}(b,\text{independentContinuant},t) \wedge \neg \text{instanceOf}(b,\text{spatialRegion},t) \\ &\quad \wedge \text{instanceOf}(t,\text{temporalRegion},t)) \end{aligned}$$

At all times t, there's a part of t when c occupies spatial region r iff every part of c occupies a part of r, and there isn't a smaller part of r that c occupies. [grv-1]

$$\begin{aligned} &\forall c,r,t (\text{instanceOf}(c,\text{independentContinuant},t) \wedge \neg \text{instanceOf}(c,\text{spatialRegion},t) \\ &\quad \wedge \text{instanceOf}(r,\text{spatialRegion},t) \\ &\quad \rightarrow \exists t_2 (\text{temporalPartOf}(t_2,t) \\ &\quad \wedge (\text{occupiesSpatialRegion}(c,r,t_2) \\ &\quad \leftrightarrow (\forall cp (\text{continuantPartOf}(cp,c,t_2) \\ &\quad \rightarrow \forall rp (\text{occupiesSpatialRegion}(cp,rp,t_2) \\ &\quad \rightarrow \text{continuantPartOf}(rp,r,t_2)))))) \\ &\quad \wedge \neg (\exists r' (r' \neq r \wedge \text{continuantPartOf}(r',r,t_2) \\ &\quad \wedge \text{occupiesSpatialRegion}(c,r',t_2)))))) \end{aligned}$$
